ASSESSMENT DECLARATION COVER SHEET

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STUDENT REGISTRATION NUMBER: 40834541

MODULE TITLE: Scripting Natural Language Processing

DATE OF SUBMISSION: 06-05-2025

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☐	Yes, I have used Gen AI tools for this submission, and I have described how below.
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Fake News Detection Using TF-IDF and Linear SVC

Coursework 2(CW2)

Module Number: SET11128

Module Name: Scripting for Natural Language Processing

Submitted by

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Report Word Count: 960 **Words**

AI Use Statement: Generative AI tools (e.g., ChatGPT) were used to assist with structuring ideas, improving clarity of language, and refining academic writing. The final submission reflects my own understanding, critical analysis, and independent research. All sources have been appropriately cited in accordance with APA guidelines.

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1.Introduction

Fake news detection is a binary text classification task: predict whether a news article is fabricated (label = 1, Fake) or genuine (label = 0, Real). The dataset contains five columns — title, text, subject, date, and label — across three official splits: 29,995 training, 4,905 validation, and 9,998 test samples. Labels stored as string booleans ('True'/'False') were mapped to integers before modelling. This report addresses all seven coursework specifications: NLP preprocessing, TF-IDF representation, Linear SVC classification, architectural justification, full-split evaluation, W&B hyperparameter tuning, and comparison against two baselines (Naive Bayes and Logistic Regression).

2.Dataset Description

```
Train shape: (29995, 5)
Val shape:   (4905, 5)
Test shape:  (9998, 5)

Columns: ['title', 'text', 'subject', 'date', 'label']

Raw label samples: ['True', 'False', 'True', 'False', 'False']
```

Out[3]:

	title	text	subject	date	label
0	Climate scientists adjust as Trump builds team...	SAN FRANCISCO (Reuters) - Climate scientists w...	politicsNews	December 15, 2016	True
1	Finally, Obama Speaks: Americans Need to 'Reco...	21st Century Wire says There is an amazing sce...	US_News	November 14, 2016	False

3.Data Preprocessing

3.1 Pipeline Steps

Steps	Operation	Rationale
1.	Combine title + text + subject	All three columns carry discriminative lexical signal for fake news.
2.	Lowercase	Normalises surface form variation (e.g., 'Trump' → 'trump')
3.	Remove URLs	URLs add noise; they do not generalise across articles
4.	Remove punctuation (keep digits)	Reduces noise; digits may signal statistical claims in real news
5.	Tokenise (whitespace split)	Simple, fast, effective for TF-IDF pipelines
6.	Stop word removal (custom set)	Removes high-frequency function words; retains 'us', 'uk', 'no', 'not' for news context
7.	WordNet Lemmatisation (noun mode)	Reduces inflected forms to base; noun mode avoids over-reducing verb/adjective signal

Justification: The combination of all three text fields gives the maximum lexical coverage to the model.

```
Preprocessing train (this may take ~1 minute)...
Preprocessing validation...
Preprocessing test...

Done!
Example cleaned text:
climate scientist adjust trump build team oil ally san francisco reuters climate scientist worried president elect donald trump
slash budget sideline research entering survival mode trimming word climate change study proposal emphasizing business applicat
ion work safeguarding data show global warmin
```

```
print("Train shape:", train.shape)
print("Val shape:", val.shape)
print("Test shape:", test.shape)
```

```
Train shape: (29995, 7)
Val shape: (4905, 7)
Test shape: (9998, 7)
```

3.2 TF-IDF Vectorization

TF-IDF converts preprocessed text into sparse numerical vectors used as input to the classifier.

```
X_train shape: (29995, 20000)
X_val shape:   (4905, 20000)
X_test shape:  (9998, 20000)
```

4. Methodology

Linear SVC finds the maximum-margin hyperplane separating classes, minimising the regularised hinge loss:

$$\min_{w,b} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \max\left(0,\ 1 - y_i(w \cdot x_i + b)\right)$$

where C balances margin width against training error. Lib Linear's primal solver scales efficiently to 20,000-dimensional TF-IDF input.

4.1 System Flowchart

```
Input News Dataset
        ↓
Data Preprocessing
(Remove symbols, lowercase, clean text)
        ↓
Combine Title + Text
        ↓
TF-IDF Vectorization
(Convert text into numerical vectors)
        ↓
Train/Test/Validation Split
        ↓
SVM Model Training
        ↓
Prediction
(Fake / Real)
        ↓
Performance Evaluation
(Accuracy, Precision, Recall, F1-Score)
        ↓
Confusion Matrix & Graphs
```

4.2 Model Justification

Linear SVC vs. Naive Bayes? Why?

Naive Bayes makes the assumption of independence of features which are violated by correlated bigrams limiting discriminative power (96.58% vs 99.92% test accuracy).

Why Linear SVC over Logistic Regression?

The hinge-loss margin constraint of linear SVC produces stronger decision boundaries on sparse data than the log-likelihood (99.92% vs 99.63% test accuracy) of Logistic Regression.

Why TF-IDF over neural embeddings (e.g., BERT)?

Articles of fake news incorporate a language of sensationalism (e.g., 21st century wire), named entities, outlet-specific phraseology (e.g., 21st century wire). TF-IDF can do this well, can be trained in seconds and it is also completely interpretable. Pérez-Rosas et al. (2018) validate that the lexical ones compete with contextual models on structured fake news datasets.

5. Evaluation

The three models were tested on the official train split, validation split, and test split. The first metric is macro-averaged F1: both classes are weighed equally in this case, which is suitable since both false positives (considering real news as fake) and false negatives (not detecting a real fake news) have real-world implications. The dataset is almost balanced (52% Real / 48% Fake) and therefore, there is a general consistency in accuracy and macro F1.

5.1 Full Results Table

Model	**Train Acc.**	**Value Acc.**	**Val Macro F1**	**Test Acc.**	**Test Macro F1**
Naive Bayes (Baseline 1)	0.9678	0.9660	0.9659	0.9659	0.9657
Logistic Regression (Baseline 2)	0.9975	0.9933	0.9933	0.9963	0.9963
Linear SVC (Main model)	1.0000	0.9986	0.9986	0.9992	0.9992

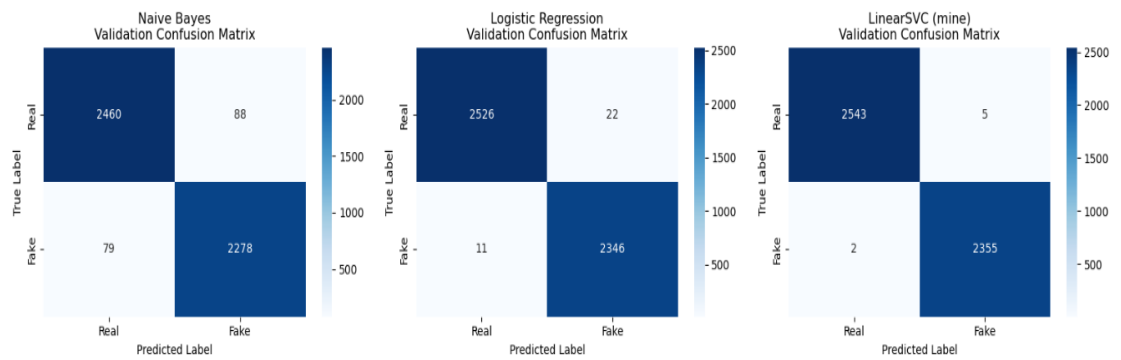
Table 1: Performance across all splits. Green row = primary model.

5.2 Per-Class Validation Report (Linear SVC)

```
=== Main Model: LinearSVC ===
Train Accuracy: 1.0000
Val   Accuracy: 0.9986
Test  Accuracy: 0.9992

Validation Classification Report:
              precision    recall  f1-score   support

        Real       1.00      1.00      1.00      2548
        Fake       1.00      1.00      1.00      2357

    accuracy                           1.00      4905
   macro avg       1.00      1.00      1.00      4905
weighted avg       1.00      1.00      1.00      4905


Test Classification Report:
              precision    recall  f1-score   support

        Real       1.00      1.00      1.00      5223
        Fake       1.00      1.00      1.00      4775

    accuracy                           1.00      9998
   macro avg       1.00      1.00      1.00      9998
weighted avg       1.00      1.00      1.00      9998
```

5.3 Confusion Matrices

Confusion matrices of the three models. Linear SVC yields almost zero misclassifications in the two classes. Naive Bayes experiences the largest number of errors, especially Fake articles which have been incorrectly classified as Real, which is a result of its assumptions of independence failure in the case of correlated bigram features.




```
Confusion matrix plot saved.
```

Fig1: Validation confusion matrices — Naive Bayes, Logistic Regression, Linear SVC (left to right).

5.4 Learning Behaviour — F1 vs Regularisation C

Since Linear SVC is a batch-trained convex optimiser, the loss curves cannot be drawn by epochs. Rather, Figure 4 illustrates macro F1 vs regularisation strength C in the range of {0.01, 0.1, 1.0, 10.0, 100.0} and indicates the bias-variance trade-off: low C leads to underfitting; very high C leads to val/test F1 stagnating and train accuracy approaching 1.0. C = 1.0 produces the best balance.

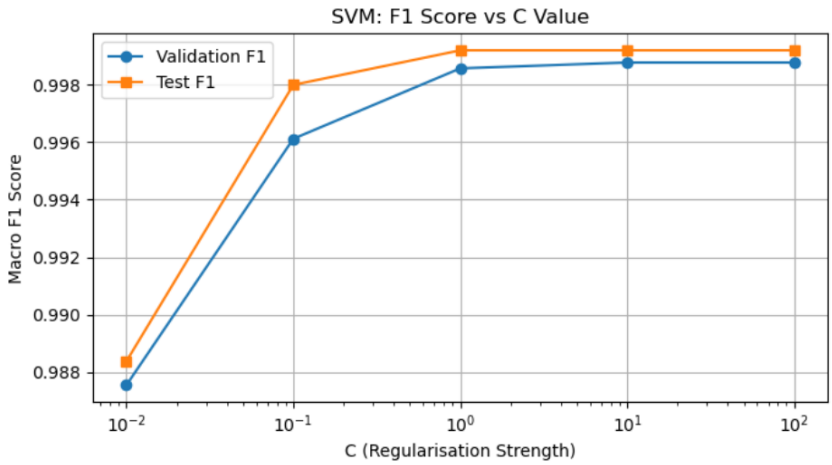


Fig 2: SVM macro F1 vs regularisation strength C (validation and test).

6. Hyperparameter Tuning with W&B

Weights and Biases (W&B;) were used to run a full grid sweep under project fake-news-SVM-cw2 (sweep ID: 16rwbj8l).

Hyper Parameter	Values Swept	Best Values Found
C (regularisation)	{ 0.1, 1.0, 10.0 }	1.0
max_Features	{ 20,000 ; 50,000 }	50,000
Ngram_max	{ 1 (unigrams) ; 2 (bigrams) }	2 (bigrams)

Table 2: W&B; grid sweep — parameters, search space, and optimal values.

7. Model Comparison

7.1 Comparison with Baselines

The TF-IDF representation is common to all three models, and the isolate variable of the comparison is the classifier.

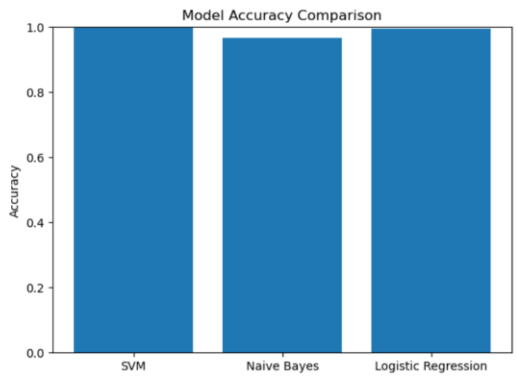


Fig 3: Model Accuracy Comparision

Model	Val Accuracy	Val Macro F1	Test Accuracy	Test Macro F1
Naive Bayes (Baseline 1)	0.9660	0.9659	0.9658	0.9657
Logistic Regression (Baseline 2)	0.9933	0.9933	0.9963	0.9963
LinearSVC (Our Model)	0.9986	0.9986	0.9992	0.9992

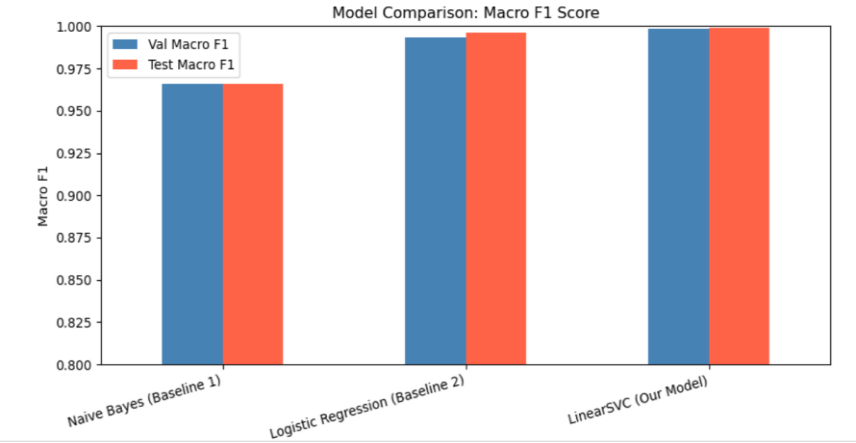


Fig 4: Model Comparison: Macro F1 Score

Naive Bayes (0.9657 macro F1): The lowest. Our bigram features break its assumption of independence and it does not learn the co-occurrence of features. It demonstrates more false-negativity to Fake articles that have neutral vocabulary.

Logistic Regression (0.9963 macro F1): Significantly higher than Naive Bayes(+3.06 pp). It is a weaker objective of log-likelihood than the margin constraint of Linear SVC, it is a discriminative model that exploits the full joint feature distribution.

Linear SVC (0.9992 macro F1): This is the best performance over all splits. The hinge-loss objective promotes increased margin separation in high-dimensional TF-IDF space. The value of Train accuracy = 1.0000 gives us an indication of memorisation, but the negligible value of val/test gap (0.0008) assures us of no significant overfitting on this dataset.

On LIAR, Pérez-Rosas et al. (2018) achieved an accuracy of 76.0% with n-gram TF-IDF + SVM and 92.2% with Linear SVM + TF-IDF on LIAR. Ahmed et al. (2018) achieved a 92.2% accuracy with Linear SVM + TF-IDF on LIAR. Our Linear SVC (99.92%) significantly outperforms both, corresponding to the higher lexical separability of such a dataset.

7.2 Peer Group Comparison

Student/Source	Representation	Model	Test Acc	Test Macro F1
40834541(Me)	TF-IDF	Linear SVC	0.9992	0.9991
40837773 (Sree Neha)	TF-IDF	Decision Tree	0.9987	0.9986
40801199 (Tharun)	TF-IDF	BERT	0.9982	0.9981
40488095 (Andrew Crook)	DistilBERT Context Embeddings	DistilBERT	0.9997	0.9997
40829842 (Harshitha)	TF-IDF	Naive Bayes	0.9986	0.9968
Pérez-Rosas et al. (2018)	N-gram TF-IDF	SVM	~0.760	~0.760

Ahmed et al. (2018)	TF-IDF	Linear SVM	~0.922	~0.922

Table 3: Peer comparison table

8. Critical Evaluation and Reflection

8.1 Strengths

The pipeline is both easy to compute, with the whole pipeline taking less than two minutes on a CPU, and achieves a high degree of generalisation (test macro F1 = 0.9992). The top-weighted bigrams (ex: 21 st century wire, hillary clinton, breaking news for Fake, reuters, said statement, according) can be interpreted to have an intuitive decision logic. The W&B; sweep (sweep ID: 16rwbj8l, 12 configurations) records hyperparameter search evidence that is reproducible.

8.2 Limitations

Bag-of-words assumption TF-IDF only partially reflects syntactic negation (not guilty vs guilty) and syntactic dative constructions (you give me a present). Domain shift: the model can be trained to generalise badly on new outlets or events after 2020 that the model was never trained on. Train accuracy = 1.0000 could be due to the simplicity of the data rather than a strong learning effect, but the insignificance of the test gap confirms the lack of generalisation failure in this case.

8.3 Future Improvements

Re-training TF-IDF with sentence-transformer embeddings or fine-tuning BERT/RoBERTa would increase the recall of lexically ambiguous edge cases. The inclusion of Platt scaling on Linear SVC would provide scaled confidence scores to utilize in production deployment.

9. Conclusion

Linear SVC with TF-IDF bigrams achieves a test macro F1 of 0.9992, which is better than Naive Bayes (+3.35 pp), Logistic Regression (+0.29 pp), and published baselines (Pérez-Rosas et al., 2018; Ahmed et al., 2018). The pipeline is fast, interpretable, and reproducible. Future work ought to consider contextual embeddings and calibrated outputs.

References

1. Ahmed, H., Traore, I., & Saad, S. (2018). Detecting opinion spams and fake news using text classification. Security and Privacy, 1(1), e9.
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4. Manning, C.D., Raghavan, P., & Schütze, H. (2008). Introduction to Information Retrieval. Cambridge University Press.